

MANUFACTURING TOOL TECHNOLOGY
SEMESTER -3(Mechanical/Metallurgy)
Week # 01

Safety standards Practiced in the machine Shop:

i. Wear Protective Gear:

Safety glasses, ear protection, gloves, safety helmet, safety shoes, and flame-resistant clothing.

ii. Machine and Electrical Safety:

- Use guards on machines and regularly check and maintain them.
- Turn off and secure machines when fixing them.
- Ground electrical equipment, use circuit breakers, cords and replace damaged.

iii. Keep the Area Safe:

- Keep the workspace clean and organized.
- Ensure good lighting, mark danger zones, and keep walkways and exits clear.

iv. Chemical and Fire Safety:

- Properly label and store chemicals and use ventilation for fumes.
- Keep fire extinguishers nearby and train everyone on how to use them.
- Store flammable materials safely.

v. Training, Procedures, and Emergency Plans:

- Conduct regular safety training and follow standard procedures.
- Report unsafe conditions and perform safety audits.
- Keep a first aid kit, train in first aid and practice evacuation plans.

Importance of housekeeping:

i. Safety:

- Prevents accidents and injuries from tools.
- Reduces fire risks.

ii. Efficiency:

- Makes it easier to find tools and materials.
- Keeps work flowing smoothly and meets regulatory standards.

iii. Machine Care:

- Keeps machines and tools in good condition.
- Prevents damage from dust and debris (flying particles).

iv. Cleanliness:

- Keeps products clean and free from contamination.
- Creates a nicer, more professional workplace for employees and visitors.



v. Health and Environmental Impact:

- Reduces dust and chemicals that can cause health issues.
- Ensures proper disposal and recycling, keeping pollutants contained

Good Shop Floor Practices (5S):

What is 5S?

- 5S is a method for organizing a workspace to make it cleaner, safer, and more efficient.
- Each "S" represents a step:

i. **Sort (Seiri):** Remove unnecessary items from the workspace.

ii. **Set in Order (Seiton):** Organize tools and materials so they are easy to find and use.

iii. **Shine (Seiso):** Clean the workspace thoroughly to spot issues early.

iv. **Standardize (Seiketsu):** Establish rules and procedures for maintaining cleanliness and organization.

v. **Sustain (Shitsuku):** Make 5S habit and continuously improve.



Importance of 5S on the Shop Floor:

1. **Efficiency:** Reduces time wasted looking for tools and materials.
2. **Safety:** Creates a safer environment by removing hazards or danger.
3. **Quality:** Improves product quality by reducing errors and defects.
4. **Morale:** Boosts employee satisfaction and pride in their work environment.
5. **Cost Savings:** Reduces waste and maintenance costs.

Understand the Principles of First Aid:

i. **Save Lives:** Priority is to keep the person alive.

ii. **Avoid Making Things Worse:** Prevent injuries from getting worse.

iii. **Promote Recovery:** Help the person heal and feel better.

iv. **Protect:** Keep them safe from further harm.

v. **Get Help:** Call emergency services if needed.

vi. **Comfort:** Offer comfort and support.



4 Essential Steps of First Aid:

Assess, Plan, Implement, Evaluate

01 Assess

Ensure safety, assess the situation, check for consciousness, breathing, and bleeding.



02 Plan

Call emergency services, decide what assistance is required, perform CPR, stop bleeding, or provide psychological aid.



03 Implement

Administer first aid, check for safety, continue monitoring the situation, and reassess the condition of the person.



04 Evaluate

Review the effectiveness of the first aid provided, repeat previous steps as necessary, and decide whether to continue or wait for emergency services.



ALLS First Aid Training

Preventive measures to be taken during Fire and Electrical emergency :

Fire Emergency:

- **Install Smoke Alarms:** Have working smoke alarms in your home.
- **Keep Fire Extinguishers:** Store fire extinguishers where it is easy to reach.
- **Plan Escape Routes:** Know how to exit quickly in a fire.
- **Cheek Machines:** Regularly inspect machines and appliances.

Electrical Emergency:

- **Turn Off Power:** If safe, turn off the power during an electrical problem.
- **Use Fire Extinguisher:** Use a fire extinguisher made for electrical fires, not water.
- **Call for Help:** Dial emergency services if needed.
- **Avoid Overloading:** Don't overload outlets or cords.

Types of Fire extinguisher (Class A, B, C, D & K):

i. **Class A Fire Extinguisher:** Used to extinguish fires caused due to combustion of cloth, wood, rubber, paper and plastics.

ii. **Class B Fire Extinguisher:** Used to extinguish fires caused due to combustion gasoline, grease and oil.

iii. **Class C Fire Extinguisher:** Used to extinguish fires caused due to electrical equipment and appliances.

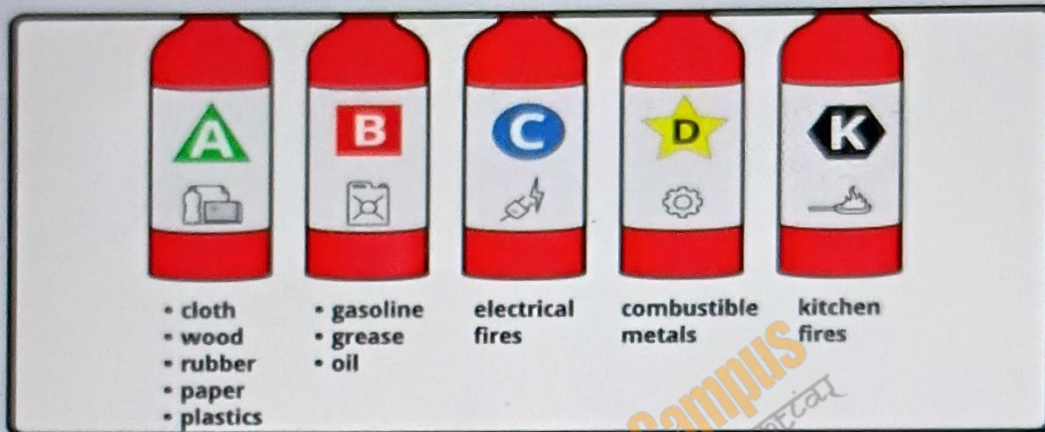
iv. **Class D Fire Extinguisher:** Used to extinguish fires caused due to combustion metals like magnesium and sodium.

v. **Class K Fire Extinguisher:** Used to extinguish fires caused due to combustion of cooking oils and fats, typically in kitchens.



Visual Watermark

Types of Fire Extinguishers



Knowledge about Safety and Environment Regulations:

- i. **Workplace Safety:** Ensures safe working conditions with required safety training and equipment.
- ii. **Environmental Protection:** Controls pollution and protects air and water quality through regulations.
- iii. **Chemical Safety:** Requires clear labelling and information on hazardous chemicals in the workplace..
- iv. **Emergency Planning:** Mandates reporting and planning for handling hazardous substances during emergencies.
- v. **Waste Management:** Regulates the proper disposal and handling of industrial waste to protect the environment.